

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 50

**Duration :60 Mins
On Class
Total Marks:50**

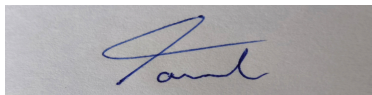
Annexure A

Viva Voce

Code of Conduct:

*I Tamil Elakiya S (192524443) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



Annexure A

Viva Voce

- Q1.What is syntax analysis?
- Q2.Why is syntax analysis needed in a compiler?
- Q3.What is the role of a parser?
- Q4.What is a Context-Free Grammar (CFG)?
- Q5.What are the components of a Context-Free Grammar?
- Q6.What is a terminal symbol?
- Q7.What is a non-terminal symbol?
- Q8.What is a production rule?
- Q9.What is the start symbol in a grammar?
- Q10.What is derivation in Context-Free Grammar?
- Q11.What is leftmost derivation?
- Q12.What is rightmost derivation?
- Q13.What is a parse tree?
- Q14.What is the difference between a parse tree and a derivation?
- Q15.What is an ambiguous grammar?
- Q16.How can ambiguity be removed from a grammar?
- Q17.What is left recursion?
- Q18.Why should left recursion be eliminated?
- Q19.What is indirect left recursion?
- Q20.What is left factoring?
- Q21.Why is left factoring required?
- Q22.What is Top-Down Parsing?
- Q23.What are the advantages of Top-Down Parsing?
- Q24.What are the limitations of Top-Down Parsing?
- Q25.What is Recursive Descent Parsing?
- Q26.What are the advantages of Recursive Descent Parsing?
- Q27.What are the limitations of Recursive Descent Parsing?
- Q28.What is backtracking in Recursive Descent Parsing?
- Q29.What is Predictive Parsing?
- Q30.Why is Predictive Parsing called an LL(1) parser?
- Q31.What is FIRST()?
- Q32.What is FOLLOW()?
- Q33.Why are FIRST and FOLLOW sets required?

- Q34. What is a Predictive Parsing Table?
- Q35. What are the conditions for constructing an LL(1) grammar?
- Q36. What is Bottom-Up Parsing?
- Q37. What is Shift-Reduce Parsing?
- Q38. What are the four actions of a Shift-Reduce Parser?
- Q39. What is a handle?
- Q40. What is handle pruning?
- Q41. What is Operator Precedence Parsing?
- Q42. What are the advantages of Operator Precedence Parsing?
- Q43. What is an LR parser?
- Q44. What are the advantages of LR parsing?
- Q45. What is an SLR parser?
- Q46. What is an augmented grammar?
- Q47. What is a viable prefix?
- Q48. What is syntax error recovery?
- Q49. What are the common syntax error recovery techniques?
- Q50. What is YACC, and why is it used in compiler design

